

## Full length article

# A coupled DQ-Heaviside-NURBS approach to investigate nonlinear dynamic response of GRE cylindrical shells under impulse loads

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## ABSTRACT

As a first endeavour, nonlinear dynamic response of the glass fibre-reinforced epoxy (GRE) laminated composite cylindrical shells under an impulse load is investigated based on the first-order shear deformation theory (FSDT) of shells. Green's strain and von Kármán hypothesis are assumed to consider the geometrical nonlinearity due to large deformation in the model. A new solution procedure composed of the differential quadrature method (DQM) based on the direct projection of the Heaviside function and a non-uniform rational B-spline (NURBS) based multi-step time integration scheme is employed to discretize the governing equations in the spatial and temporal domains, respectively. The approach is validated by showing its fast convergence rate and performing comparison studies with available solutions in the limited available cases. A comprehensive parametric study is performed then on the model and the effects of the geometrical parameters, number of layers, load location, time durations, and types of impulse loading on the nonlinear dynamic responses of GRE laminated composite cylindrical shells are investigated.

## 1. Introduction

Glass fibre-reinforced epoxy (GRE) composite cylindrical shells are widely used in a wide range of applications in different industrial sectors such as oil and gas transportation pipes, petro-chemical structures, oil and gas industries, etc [1,2]. Compared with conventional metallic materials, GRE composites represent superior performance such as high specific strength and strength; provide significant weight reduction [2,3], and reveal outstanding mechanical properties such as strength, corrosion and wear resistance [4]. Thanks to this high corrosion and wear resistance, the GRE-based structures show fewer problems and have lower maintenance costs with shorter repair time than the metal ones. Nonlinear analysis of GRE shell structures has become increasingly essential in the study of structural response to dangerous loads such as impulse. The aforementioned advantages well justify research on the nonlinear dynamic behaviour of GRE laminated composite cylindrical shells.

Linear and nonlinear dynamics of functionally graded (FG) and laminated composite shells, plates and beams are widely investigated in the literatures [5–48]. Among them, there is also an specific focus and concern on the dynamic response, instability and buckling analyses

of the composite cylindrical shells under impulse load [5,6,9,19,35,42,48]. In addition, some researchers have simulated the responses of the GRE composite structures using the experimental and numerical approaches and some of the contributions in the field are reviewed in the continuation.

Elghazouli et al. [49] studied the buckling behaviour of laminated composite cylinders made from glass fibre reinforced plastic (GFRP) experimentally. They demonstrated the effect of laminate orientation on the buckling strength of composite cylinders. Zaretsky et al. [50] investigated the dynamic response of a woven glass fibres reinforced epoxy composite to a planar impact loading. Deniz et al. [51] applied an experimental method to investigate the effect of seawater on the failure pressure and impact behaviours of polymer–matrix composites (PMCs). They used six layers with a  $\pm 55^\circ$  winding angle consist of E-glass fibre and epoxy resin to fabricate the composite pipes. Abdul Majid et al. [52] presented the modelling of the stress–strain response of GRE composite pipes subjected to multiaxial loadings. Perillo et al. [53] employed an advanced 3-D finite element (FE) model and experimental test to investigate the impact effects on the GRE pipes, produced by filament winding. Arikan and Sayman [54] investigated the effect of

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resin type on the impact response of GRE composites subjected to the low velocity single and repeated impacts. They fabricated GRE composites with two types of resin, polypropylene, and epoxy. Jayarami Reddy et al. [55] experimentally studied the response of the H- glass/epoxy composite laminates against explosive blast loading. The prediction of the erosion behaviour of GRE composites subjected to solid particulate impacts was studied by Agarwal et al. [56] using a finite element method model. Das et al. [57] numerically and experimentally simulated the free vibration of the cracked woven fiberglass/epoxy composite beam. They could utilize finite element software ANSYS as a numerical technique to explore the free vibration and to validate the experiment.

Having the aforementioned overview on the studies within the field, and to the best of the authors' knowledge, there is still no research available on the nonlinear dynamic analysis of GRE laminated composite cylindrical shells under impulse loading. The practical and scientific importance of the subject is the main motivation for detailed investigation in this study. This is despite the fact that there is a high chance of occurrence of (near) impulse loading on shell-like structures and also impulsive loadings are widely used in structural health monitoring [58–60].

To model the nonlinear dynamic behaviour of GRE laminated composite cylindrical shells under impulse loading, the case study, here is assumed to be a multi-layer composite structure. Epoxy is combined with glass fibres to build up each composite layer. Hamilton's principle and FSDT theory of shells are employed to derive the nonlinear equations of motion and corresponding boundary conditions of the GRE composite cylindrical shells. The direct projection of the Heaviside function is adopted to simulate the variation of the field variables at the coordinates of the load in both axial and circumferential directions. Accordingly, a differential quadrature method (DQM) [61–71] based on the direct projection of the Heaviside function [72] is used to discretize the governing equations in the spatial domain which yields a non-oscillatory approximation of the solution. Moreover, this new approach is shown to be superior over other methods such as the DQMR approach (DQM with regularized Dirac-delta function) in terms of both accuracy and convergence [72]. Then, the obtained system of nonlinear ordinary equations is solved using a multi-step NURBS-based technique [64,69] in conjunction with the Newton–Raphson method. Comparative studies between the multi-step technique based on the NURBS curves and Newmark approach demonstrate that the required time step of this multi-step method is much less than that of the well-known Newmark technique. However, in the current work, a new DQM based on direct projection of the Heaviside function in conjunction with the NURBS based multi-step technique are used to solve the nonlinear dynamic response of GRE laminated composite cylindrical shells under an impulse load. Convergence study and comparing these results with available analytical solutions of the cases with limited and simpler conditions verify the accuracy of the proposed procedure. The effects of different parameters like geometries of the shell (parameters such as thickness and length-to-mean radius ratios), the number of layers, time durations, and also types of loads along with load locations are also studied to a wider picture of their impacts on the nonlinear behaviour of the GRE laminated composite cylindrical shells.

## 2. Governing equations

The geometry and coordinate system of the GRE cylindrical shell are shown in Fig. 1. This shell is composed of  $N_e$  perfectly bonded GRE reinforced composite layers subjected to impulse loading. The GRE composite shell has, respectively, winding angle  $\varphi$ , inner radius  $R_i$  and outer radius  $R_o$ . The thickness of the GRE layer and total thickness are defined as  $h_c$  and  $h$ , respectively. As depicted in Fig. 1(a), a cylindrical coordinate system with the coordinate variables  $(x, \theta, z)$  is chosen to identify the material points of the shell in the undeformed configuration.

As stated by FSDT, the displacement field of an arbitrary material point of the  $e$ th GRE laminated composite shell is expressed as

$$\begin{aligned}\bar{u}(x, \theta, z, t) &= u(x, \theta, t) + z\psi^x(x, \theta, t), \\ \bar{v}(x, \theta, z, t) &= v(x, \theta, t) + z\psi^\theta(x, \theta, t), \quad \bar{w}(x, \theta, z, t) = w(x, \theta, t)\end{aligned}\quad (1a-c)$$

where  $u$ ,  $v$  and  $w$  are the displacement components of an arbitrary material point in the mid-plane of the shell along the  $x$ ,  $\theta$  and  $z$  directions, respectively;  $\psi^x, \psi^\theta$  represent the transverse normal about the  $\theta$  and  $x$  axis, respectively.

Using Hamilton's principle Eq. (2), the nonlinear equations of motion and the related boundary conditions of the GRE composite shell under an impulse load are derived [25]

$$\int_{t_1}^{t_2} (\delta K - \delta U + \delta W) dt = 0 \quad (2)$$

where  $t_1$  and  $t_2$  indicate two arbitrary times;  $\delta W$  is the virtual work of the external load;  $\delta K$  and  $\delta U$  represent the variation of the kinetic energy and potential energy of the GRE composite shell, respectively, which can be expressed as

$$\begin{aligned}\delta U &= \int_0^{2\pi} \int_0^L \int_{-\frac{h}{2}}^{\frac{h}{2}} [\sigma_{xx} (\delta \epsilon_{xx}^L + \delta \epsilon_{xx}^{NL}) + \sigma_{\theta\theta} (\delta \epsilon_{\theta\theta}^L + \delta \epsilon_{\theta\theta}^{NL}) \\ &\quad + \sigma_{zx} (\delta \gamma_{zx}^L + \delta \gamma_{zx}^{NL}) + \sigma_{\theta x} (\delta \gamma_{\theta x}^L + \delta \gamma_{\theta x}^{NL}) \\ &\quad + \sigma_{z\theta} (\delta \gamma_{z\theta}^L + \delta \gamma_{z\theta}^{NL})] (R+z) dz dx d\theta\end{aligned}\quad (3)$$

$$\delta K = \int_0^{2\pi} \int_0^L \int_{-\frac{h}{2}}^{\frac{h}{2}} \rho \left[ \frac{\partial \bar{u}}{\partial t} \frac{\partial \delta \bar{u}}{\partial t} + \frac{\partial \bar{v}}{\partial t} \frac{\partial \delta \bar{v}}{\partial t} + \frac{\partial \bar{w}}{\partial t} \frac{\partial \delta \bar{w}}{\partial t} \right] (R+z) dz dx d\theta \quad (4)$$

$$\delta W = \int_0^{2\pi} \int_0^L P_0 \phi(t) \delta(x - x_0) \delta(\theta - \theta_0) \delta \bar{w} R_o dx d\theta \quad (5)$$

where  $\sigma_{ij}$  ( $i, j = x, \theta, z$ ) are the stress tensor components in the GRE reinforced composite layers;  $\epsilon_{ij}$  and  $\gamma_{ij}$  denote the normal and shear components of Green's strain tensor, respectively, which are presented in Appendix; superscript  $L$  and  $NL$  are used to identify the linear and nonlinear terms of Green's strain tensor, respectively;  $\rho$  indicates the mass density;  $P_0$ ,  $x_0$  and  $\theta_0$  denote the value and the locations of the impulse load along the  $x$ - and  $\theta$ -directions, respectively; also,  $\phi$  is the type of the impulse loading.

Substituting Eqs. (3)–(5) into Eq. (2) and using the strain–displacement and the stress–strain relations yields the equations of motion (6)–(10) and the geometric and natural compatibility conditions (11) and (12) in terms of the displacement and the bending rotation components,

*Equations of motion:*

$\delta u$ :

$$\begin{aligned}&A_1^{11} \frac{\partial^2 u}{\partial x^2} + A_1^{16} \frac{\partial^2 v}{\partial x^2} + A_{-1}^{66} \frac{\partial^2 u}{\partial \theta^2} + (A_0^{66} + A_0^{12}) \frac{\partial^2 v}{\partial \theta \partial x} + 2A_0^{16} \frac{\partial^2 u}{\partial \theta \partial x} \\ &+ A_0^{12} \frac{\partial w}{\partial x} + B_1^{11} \frac{\partial^2 \psi^x}{\partial x^2} + B_1^{16} \frac{\partial^2 \psi^\theta}{\partial x^2} + B_{-1}^{66} \frac{\partial^2 \psi^x}{\partial \theta^2} + 2B_0^{16} \frac{\partial^2 \psi^x}{\partial \theta \partial x} \\ &+ (B_0^{66} + B_0^{12}) \frac{\partial^2 \psi^\theta}{\partial \theta \partial x} + A_{-1}^{26} \frac{\partial^2 v}{\partial \theta^2} + A_{-1}^{26} \frac{\partial^2 \psi^\theta}{\partial \theta^2} + B_{-1}^{26} \frac{\partial w}{\partial \theta} \\ &+ \frac{A_{-1}^{11}}{2} \frac{\partial}{\partial x} \left( \frac{\partial w}{\partial x} \right)^2 + \frac{A_{-1}^{12}}{2} \frac{\partial}{\partial x} \left( \frac{\partial w}{\partial \theta} \right)^2 + A_{-1}^{66} \frac{\partial}{\partial \theta} \left( \frac{\partial w}{\partial x} \frac{\partial w}{\partial \theta} \right) \\ &+ \frac{A_0^{16}}{2} \frac{\partial}{\partial \theta} \left( \frac{\partial w}{\partial x} \right)^2 + A_0^{16} \frac{\partial}{\partial x} \left( \frac{\partial w}{\partial x} \frac{\partial w}{\partial \theta} \right) = I_{11} \frac{\partial^2 u}{\partial t^2} + I_{12} \frac{\partial^2 \psi^x}{\partial t^2}\end{aligned}\quad (6)$$

$\delta v$ :

$$\begin{aligned}&(A_0^{12} + A_0^{66}) \frac{\partial^2 u}{\partial \theta \partial x} + A_1^{66} \frac{\partial^2 v}{\partial x^2} + A_{-1}^{22} \frac{\partial^2 v}{\partial \theta^2} - (A_{-1}^{44} + A_{-1}^{45}) v \\ &+ (A_{-1}^{44} + A_{-1}^{45} + A_{-1}^{22}) \frac{\partial w}{\partial \theta} + (B_0^{12} + B_0^{66}) \frac{\partial^2 \psi^x}{\partial \theta \partial x} \\ &+ B_1^{66} \frac{\partial^2 \psi^\theta}{\partial x^2} + B_{-1}^{22} \frac{\partial^2 \psi^\theta}{\partial \theta^2} + (A_0^{44} + A_0^{45} - B_{-1}^{44} - B_{-1}^{45}) \psi^\theta + (A_0^{26}) \frac{\partial w}{\partial x} \\ &+ 2A_0^{26} \frac{\partial^2 v}{\partial \theta \partial x} + 2B_0^{26} \frac{\partial^2 \psi^\theta}{\partial \theta \partial x}\end{aligned}$$

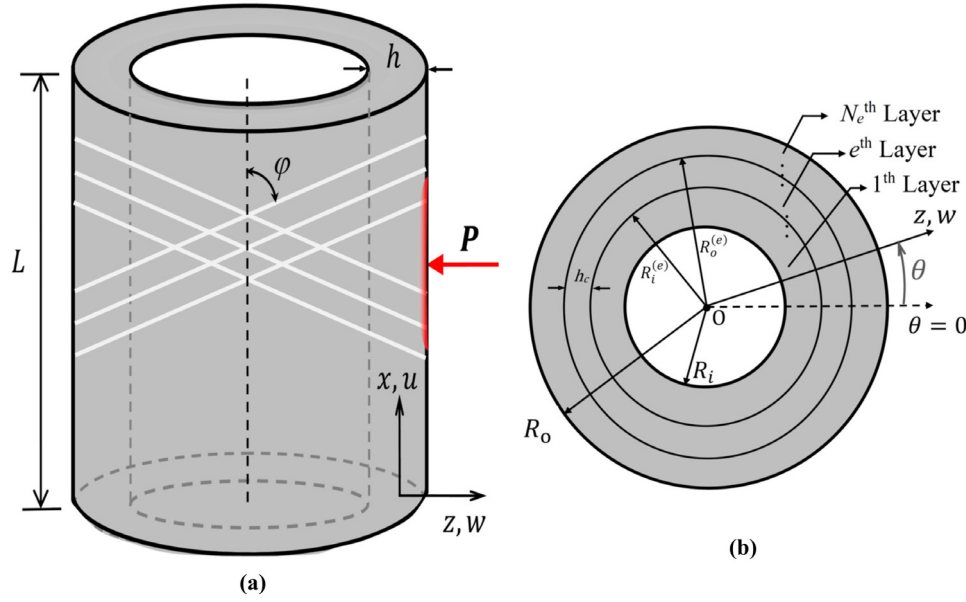


Fig. 1. Geometry of GRE laminated composite cylindrical shells.

$$\begin{aligned}
 & + B_{-1}^{26} \frac{\partial^2 \psi^x}{\partial \theta^2} + A_{-1}^{26} \frac{\partial^2 u}{\partial \theta^2} + B_{-1}^{16} \frac{\partial^2 \psi^x}{\partial x^2} + A_{-1}^{16} \frac{\partial^2 u}{\partial x^2} + \frac{A_{-2}^{22}}{2} \frac{\partial}{\partial \theta} \left( \frac{\partial w}{\partial \theta} \right)^2 \\
 & + \frac{A_0^{12}}{2} \frac{\partial}{\partial \theta} \left( \frac{\partial w}{\partial x} \right)^2 + A_0^{66} \frac{\partial}{\partial x} \left( \frac{\partial w}{\partial x} \frac{\partial w}{\partial \theta} \right) + A_{-1}^{26} \frac{\partial}{\partial \theta} \left( \frac{\partial w}{\partial x} \frac{\partial w}{\partial \theta} \right) \\
 & + \frac{A_1^{16}}{2} \frac{\partial}{\partial x} \left( \frac{\partial w}{\partial x} \right)^2 = I_{11} \frac{\partial^2 v}{\partial t^2} + I_{12} \frac{\partial^2 \psi^\theta}{\partial t^2}
 \end{aligned} \quad (7)$$

$$\begin{aligned}
 \delta w : \\
 & - A_0^{12} \frac{\partial u}{\partial x} - A_{-1}^{26} \frac{\partial u}{\partial \theta} - (A_{-1}^{22} + A_{-1}^{44} - A_{-1}^{45}) \frac{\partial v}{\partial \theta} \\
 & + (A_1^{55} + A_1^{45}) \frac{\partial^2 w}{\partial x^2} + (A_{-1}^{44} + A_{-1}^{45}) \frac{\partial^2 w}{\partial \theta^2} - A_{-1}^{22} w - A_0^{26} \frac{\partial v}{\partial x} \\
 & + (A_1^{55} - B_0^{12} + A_1^{45}) \frac{\partial \psi^x}{\partial x} - B_{-1}^{26} \frac{\partial \psi^x}{\partial \theta} \\
 & + (A_0^{44} - B_{-1}^{22} - B_{-1}^{44} + A_0^{45} - B_{-1}^{45}) \frac{\partial \psi^\theta}{\partial \theta} - B_0^{26} \frac{\partial \psi^\theta}{\partial x} \\
 & + A_1^{11} \frac{\partial}{\partial x} \left( \frac{\partial u}{\partial x} \frac{\partial w}{\partial x} \right) + B_1^{11} \frac{\partial}{\partial x} \left( \frac{\partial \psi^x}{\partial x} \frac{\partial w}{\partial x} \right) \\
 & + \frac{A_1^{11}}{2} \frac{\partial}{\partial x} \left[ \left( \frac{\partial w}{\partial x} \right)^2 \frac{\partial w}{\partial x} \right] + A_0^{12} \frac{\partial}{\partial x} \left( \frac{\partial v}{\partial \theta} \frac{\partial w}{\partial x} \right) \\
 & + B_0^{12} \frac{\partial}{\partial x} \left( \frac{\partial \psi^\theta}{\partial \theta} \frac{\partial w}{\partial x} \right) + A_0^{12} \frac{\partial}{\partial x} \left( w \frac{\partial w}{\partial x} \right) \\
 & + \frac{A_{-1}^{12}}{2} \frac{\partial}{\partial x} \left[ \left( \frac{\partial w}{\partial \theta} \right)^2 \frac{\partial w}{\partial x} \right] + \frac{A_{-2}^{22}}{2} \left( \frac{\partial w}{\partial \theta} \right)^2 + \frac{A_0^{12}}{2} \left( \frac{\partial w}{\partial x} \right)^2 \\
 & + A_{-2}^{22} \frac{\partial}{\partial \theta} \left( \frac{\partial v}{\partial \theta} \frac{\partial w}{\partial \theta} \right) + B_{-2}^{22} \frac{\partial}{\partial \theta} \left( \frac{\partial \psi^\theta}{\partial \theta} \frac{\partial w}{\partial \theta} \right) + A_{-2}^{22} \frac{\partial}{\partial \theta} \left( w \frac{\partial w}{\partial \theta} \right) \\
 & + \frac{A_{-3}^{22}}{2} \frac{\partial}{\partial \theta} \left[ \left( \frac{\partial w}{\partial \theta} \right)^2 \frac{\partial w}{\partial \theta} \right] + A_{-1}^{12} \frac{\partial}{\partial \theta} \left( \frac{\partial u}{\partial x} \frac{\partial w}{\partial \theta} \right) \\
 & + B_{-1}^{12} \frac{\partial}{\partial \theta} \left( \frac{\partial \psi^x}{\partial x} \frac{\partial w}{\partial \theta} \right) + \frac{A_{-1}^{12}}{2} \frac{\partial}{\partial \theta} \left[ \left( \frac{\partial w}{\partial x} \right)^2 \frac{\partial w}{\partial \theta} \right] \\
 & + A_{-1}^{66} \frac{\partial}{\partial x} \left[ \left( \frac{\partial w}{\partial \theta} \right)^2 \frac{\partial w}{\partial x} \right] + A_{-1}^{66} \frac{\partial}{\partial \theta} \left[ \left( \frac{\partial w}{\partial x} \right)^2 \frac{\partial w}{\partial \theta} \right] \\
 & + A_1^{16} \frac{\partial}{\partial x} \left( \frac{\partial v}{\partial x} \frac{\partial w}{\partial \theta} \right) + B_1^{16} \frac{\partial}{\partial x} \left( \frac{\partial \psi^\theta}{\partial x} \frac{\partial w}{\partial \theta} \right) \\
 & + A_0^{16} \frac{\partial}{\partial x} \left( \frac{\partial u}{\partial \theta} \frac{\partial w}{\partial x} \right) + B_0^{16} \frac{\partial}{\partial x} \left( \frac{\partial \psi^x}{\partial \theta} \frac{\partial w}{\partial x} \right) - A_{-1}^{26} \frac{\partial w}{\partial x} \frac{\partial w}{\partial \theta} \\
 & + \frac{3A_0^{16}}{2} \frac{\partial}{\partial x} \left[ \left( \frac{\partial w}{\partial x} \right)^2 \frac{\partial w}{\partial \theta} \right] + A_{-1}^{26} \frac{\partial}{\partial \theta} \left( \frac{\partial w}{\partial \theta} \frac{\partial v}{\partial x} \right) + A_{-1}^{26} \frac{\partial}{\partial \theta} \left( \frac{\partial v}{\partial \theta} \frac{\partial w}{\partial x} \right)
 \end{aligned}$$

$$\begin{aligned}
 & + A_{-2}^{26} \frac{\partial}{\partial \theta} \left( \frac{\partial u}{\partial \theta} \frac{\partial w}{\partial \theta} \right) + A_0^{16} \frac{\partial}{\partial x} \left( \frac{\partial u}{\partial x} \frac{\partial w}{\partial \theta} \right) \\
 & + B_{-2}^{26} \frac{\partial}{\partial \theta} \left( \frac{\partial \psi^x}{\partial \theta} \frac{\partial w}{\partial \theta} \right) + A_{-2}^{26} \frac{\partial}{\partial \theta} \left[ \left( \frac{\partial w}{\partial \theta} \right)^2 \frac{\partial w}{\partial x} \right] \\
 & + B_0^{16} \frac{\partial}{\partial x} \left( \frac{\partial \psi^x}{\partial x} \frac{\partial w}{\partial \theta} \right) + B_{-1}^{26} \frac{\partial}{\partial \theta} \left( \frac{\partial \psi^\theta}{\partial x} \frac{\partial w}{\partial \theta} \right) \\
 & + A_{-1}^{26} \frac{\partial}{\partial x} \left( \frac{\partial v}{\partial \theta} \frac{\partial w}{\partial \theta} \right) + B_{-1}^{26} \frac{\partial}{\partial x} \left( \frac{\partial \psi^\theta}{\partial \theta} \frac{\partial w}{\partial \theta} \right) + A_{-1}^{26} \frac{\partial}{\partial x} \left( w \frac{\partial w}{\partial \theta} \right) \\
 & + A_0^{16} \frac{\partial}{\partial \theta} \left( \frac{\partial u}{\partial x} \frac{\partial w}{\partial x} \right) + B_0^{16} \frac{\partial}{\partial \theta} \left( \frac{\partial \psi^x}{\partial x} \frac{\partial w}{\partial x} \right) \\
 & + \frac{A_0^{16}}{2} \frac{\partial}{\partial \theta} \left[ \left( \frac{\partial w}{\partial x} \right)^2 \frac{\partial w}{\partial x} \right] + B_{-1}^{26} \frac{\partial}{\partial \theta} \left( \frac{\partial \psi^\theta}{\partial \theta} \frac{\partial w}{\partial x} \right) + A_{-1}^{26} \frac{\partial}{\partial \theta} \left( w \frac{\partial w}{\partial x} \right) \\
 & = P_0 \phi(t) \frac{\partial H^x}{\partial x} \frac{\partial H^\theta}{\partial \theta} + I_{11} \frac{\partial^2 w}{\partial t^2}
 \end{aligned} \quad (8)$$

$$\begin{aligned}
 \delta \varphi^x : \\
 & B_1^{11} \frac{\partial^2 u}{\partial x^2} + B_1^{16} \frac{\partial^2 v}{\partial x^2} + B_{-1}^{66} \frac{\partial^2 u}{\partial \theta^2} + (B_0^{66} + B_0^{12}) \frac{\partial^2 v}{\partial \theta \partial x} + 2B_0^{16} \frac{\partial^2 u}{\partial \theta \partial x} \\
 & + (B_1^{12} - A_1^{55} - A_1^{45}) \frac{\partial w}{\partial x} + D_1^{11} \frac{\partial^2 \psi^x}{\partial x^2} \\
 & + D_1^{16} \frac{\partial^2 \psi^\theta}{\partial x^2} + D_{-1}^{66} \frac{\partial^2 \psi^x}{\partial \theta^2} + 2D_0^{16} \frac{\partial^2 \psi^x}{\partial \theta \partial x} + (D_0^{66} + D_0^{12}) \frac{\partial^2 \psi^\theta}{\partial \theta \partial x} \\
 & - (A_1^{55} + A_1^{45}) \psi^x + B_{-1}^{26} \frac{\partial^2 v}{\partial \theta^2} + B_{-1}^{26} \frac{\partial^2 \psi^\theta}{\partial \theta^2} \\
 & + D_{-1}^{26} \frac{\partial w}{\partial \theta} + \frac{B_{-1}^{11}}{2} \frac{\partial}{\partial x} \left( \frac{\partial w}{\partial x} \right)^2 + \frac{B_{-1}^{12}}{2} \frac{\partial}{\partial x} \left( \frac{\partial w}{\partial \theta} \right)^2 \\
 & + B_{-1}^{66} \frac{\partial}{\partial \theta} \left( \frac{\partial w}{\partial x} \frac{\partial w}{\partial \theta} \right) + \frac{B_0^{16}}{2} \frac{\partial}{\partial \theta} \left( \frac{\partial w}{\partial x} \right)^2 + B_0^{16} \frac{\partial}{\partial x} \left( \frac{\partial w}{\partial x} \frac{\partial w}{\partial \theta} \right) \\
 & = I_{11} \frac{\partial^2 u}{\partial t^2} + I_{12} \frac{\partial^2 \psi^x}{\partial t^2}
 \end{aligned} \quad (9)$$

$$\begin{aligned}
 \delta \varphi^\theta : \\
 & (B_0^{12} + B_0^{66}) \frac{\partial^2 u}{\partial \theta \partial x} + B_1^{66} \frac{\partial^2 v}{\partial x^2} + B_{-1}^{22} \frac{\partial^2 v}{\partial \theta^2} \\
 & - (B_{-1}^{44} + A_0^{44} + B_{-1}^{45} + A_0^{45}) v + D_1^{66} \frac{\partial^2 \psi^\theta}{\partial x^2} + D_{-1}^{22} \frac{\partial^2 \psi^\theta}{\partial \theta^2} \\
 & + (B_{-1}^{44} - A_0^{44} + B_{-1}^{45} + B_{-1}^{22} - A_0^{45}) \frac{\partial w}{\partial \theta} \\
 & + (B_0^{44} - A_1^{44} + B_0^{45} - D_{-1}^{44} + B_0^{44} - D_{-1}^{45} - A_1^{45} + B_0^{45}) \varphi^\theta + B_0^{26} \frac{\partial w}{\partial x}
 \end{aligned}$$

$$\begin{aligned}
& + (D_0^{12} + D_0^{66}) \frac{\partial^2 \psi^x}{\partial \theta \partial x} + 2B_0^{26} \frac{\partial^2 v}{\partial \theta \partial x} + 2D_0^{26} \frac{\partial^2 \psi^\theta}{\partial \theta \partial x} + D_{-1}^{26} \frac{\partial^2 \psi^x}{\partial \theta^2} \\
& + B_{-1}^{26} \frac{\partial^2 u}{\partial \theta^2} + D_{-1}^{16} \frac{\partial^2 \psi^x}{\partial x^2} + B_{-1}^{16} \frac{\partial^2 u}{\partial x^2} \\
& + \frac{B_{-2}^{22}}{2} \frac{\partial}{\partial \theta} \left( \frac{\partial w}{\partial \theta} \right)^2 + \frac{B_0^{12}}{2} \frac{\partial}{\partial \theta} \left( \frac{\partial w}{\partial x} \right)^2 \\
& + B_0^{66} \frac{\partial}{\partial x} \left( \frac{\partial w}{\partial x} \frac{\partial w}{\partial \theta} \right) + B_{-1}^{26} \frac{\partial}{\partial \theta} \left( \frac{\partial w}{\partial x} \frac{\partial w}{\partial \theta} \right) \\
& + \frac{B_1^{16}}{2} \frac{\partial}{\partial x} \left( \frac{\partial w}{\partial x} \right)^2 = I_{11} \frac{\partial^2 v}{\partial t^2} + I_{12} \frac{\partial^2 \psi^\theta}{\partial t^2}
\end{aligned} \quad (10)$$

Geometric and natural compatibility conditions at  $\theta = 0$  and  $\theta = 2\pi$ :

$$u(x, 0, z, t) = u(x, 2\pi, z, t), v(x, 0, z, t) = v(x, 2\pi, z, t),$$

$$w(x, 0, z, t) = w(x, 2\pi, z, t), \psi^x(x, 0, z, t) = \psi^x(x, 2\pi, z, t),$$

$$\psi^\theta(x, 0, z, t) = \psi^\theta(x, 2\pi, z, t) \text{ and}$$

$$\begin{aligned}
& \left[ A_0^{16} \frac{\partial u}{\partial x} + A_{-1}^{26} \frac{\partial v}{\partial \theta} + A_{-1}^{26} w + B_0^{16} \frac{\partial \psi^x}{\partial x} + B_{-1}^{26} \frac{\partial \psi^\theta}{\partial \theta} \right. \\
& \left. + \frac{A_0^{16}}{2} \left( \frac{\partial w}{\partial x} \right)^2 + A_{-1}^{66} \frac{\partial w}{\partial x} \frac{\partial w}{\partial \theta} \right] \Big|_{\theta=0} = \\
& \left[ A_0^{16} \frac{\partial u}{\partial x} + A_{-1}^{26} \frac{\partial v}{\partial \theta} + A_{-1}^{26} w + B_0^{16} \frac{\partial \psi^x}{\partial x} \right. \\
& \left. + B_{-1}^{26} \frac{\partial \psi^\theta}{\partial \theta} + \frac{A_0^{16}}{2} \left( \frac{\partial w}{\partial x} \right)^2 + A_{-1}^{66} \frac{\partial w}{\partial x} \frac{\partial w}{\partial \theta} \right] \Big|_{\theta=2\pi}, \\
& \left[ B_0^{16} \frac{\partial u}{\partial x} + B_{-1}^{26} \frac{\partial v}{\partial \theta} + B_{-1}^{26} w + D_0^{16} \frac{\partial \psi^x}{\partial x} + D_{-1}^{26} \frac{\partial \psi^\theta}{\partial \theta} \right. \\
& \left. + \frac{B_0^{16}}{2} \left( \frac{\partial w}{\partial x} \right)^2 + B_{-1}^{66} \frac{\partial w}{\partial x} \frac{\partial w}{\partial \theta} \right] \Big|_{\theta=0} = \\
& \left[ B_0^{16} \frac{\partial u}{\partial x} + B_{-1}^{26} \frac{\partial v}{\partial \theta} + B_{-1}^{26} w + D_0^{16} \frac{\partial \psi^x}{\partial x} + D_{-1}^{26} \frac{\partial \psi^\theta}{\partial \theta} \right. \\
& \left. + \frac{B_0^{16}}{2} \left( \frac{\partial w}{\partial x} \right)^2 + B_{-1}^{66} \frac{\partial w}{\partial x} \frac{\partial w}{\partial \theta} \right] \Big|_{\theta=2\pi}, \\
& \left[ (A_0^{45} - B_{-1}^{45}) \psi^\theta + A_{-1}^{45} \frac{\partial w}{\partial \theta} - A_{-1}^{45} v + A_{-1}^{26} \frac{\partial v}{\partial x} \frac{\partial w}{\partial \theta} + B_{-1}^{26} \frac{\partial \psi^\theta}{\partial x} \frac{\partial w}{\partial \theta} \right. \\
& + A_{-2}^{26} \frac{\partial u}{\partial \theta} \frac{\partial w}{\partial \theta} + B_{-2}^{26} \frac{\partial \psi^x}{\partial \theta} \frac{\partial w}{\partial \theta} \\
& + A_{-2}^{26} \frac{\partial w}{\partial x} \left( \frac{\partial w}{\partial \theta} \right)^2 + A_0^{16} \frac{\partial u}{\partial x} \frac{\partial w}{\partial x} + B_0^{16} \frac{\partial \psi^x}{\partial x} \frac{\partial w}{\partial x} \\
& + \frac{A_0^{16}}{2} \left( \frac{\partial w}{\partial x} \right)^3 + A_{-1}^{26} \frac{\partial v}{\partial \theta} \frac{\partial w}{\partial x} + B_{-1}^{26} \frac{\partial \psi^\theta}{\partial \theta} \frac{\partial w}{\partial x} \\
& + A_{-1}^{26} w \frac{\partial w}{\partial x} + A_{-2}^{22} \frac{\partial v}{\partial \theta} \frac{\partial w}{\partial \theta} + B_{-2}^{22} \frac{\partial \psi^\theta}{\partial \theta} \frac{\partial w}{\partial \theta} + A_{-2}^{22} w \frac{\partial w}{\partial \theta} \\
& + \frac{A_{-3}^{22}}{2} \left( \frac{\partial w}{\partial \theta} \right)^3 + A_{-1}^{12} \frac{\partial u}{\partial x} \frac{\partial w}{\partial \theta} + B_{-1}^{12} \frac{\partial \psi^x}{\partial x} \frac{\partial w}{\partial \theta} \\
& + \left( \frac{A_{-1}^{12}}{2} + A_{-1}^{66} \right) \frac{\partial w}{\partial \theta} \left( \frac{\partial w}{\partial x} \right)^2 + A_0^{66} \frac{\partial v}{\partial x} \frac{\partial w}{\partial x} + B_0^{66} \frac{\partial \psi^\theta}{\partial x} \frac{\partial w}{\partial x} \\
& \left. + A_{-1}^{66} \frac{\partial u}{\partial \theta} \frac{\partial w}{\partial x} + B_{-1}^{66} \frac{\partial \psi^x}{\partial \theta} \frac{\partial w}{\partial x} \right] \Big|_{\theta=0} = \\
& \left[ (A_0^{45} - B_{-1}^{45}) \psi^\theta + A_{-1}^{45} \frac{\partial w}{\partial \theta} - A_{-1}^{45} v + A_{-1}^{26} \frac{\partial v}{\partial x} \frac{\partial w}{\partial \theta} + B_{-1}^{26} \frac{\partial \psi^\theta}{\partial x} \frac{\partial w}{\partial \theta} \right. \\
& + A_{-2}^{26} \frac{\partial u}{\partial \theta} \frac{\partial w}{\partial \theta} + B_{-2}^{26} \frac{\partial \psi^x}{\partial \theta} \frac{\partial w}{\partial \theta} \\
& + A_{-2}^{26} \frac{\partial w}{\partial x} \left( \frac{\partial w}{\partial \theta} \right)^2 + A_0^{16} \frac{\partial u}{\partial x} \frac{\partial w}{\partial x} + B_0^{16} \frac{\partial \psi^x}{\partial x} \frac{\partial w}{\partial x} \\
& + \frac{A_0^{16}}{2} \left( \frac{\partial w}{\partial x} \right)^3 + A_{-1}^{26} \frac{\partial v}{\partial \theta} \frac{\partial w}{\partial x} + B_{-1}^{26} \frac{\partial \psi^\theta}{\partial \theta} \frac{\partial w}{\partial x} \\
& \left. + A_{-1}^{26} w \frac{\partial w}{\partial x} + A_{-2}^{22} \frac{\partial v}{\partial \theta} \frac{\partial w}{\partial \theta} + B_{-2}^{22} \frac{\partial \psi^\theta}{\partial \theta} \frac{\partial w}{\partial \theta} + A_{-2}^{22} w \frac{\partial w}{\partial \theta} \right]
\end{aligned}$$

$$\begin{aligned}
& + \frac{A_{-3}^{22}}{2} \left( \frac{\partial w}{\partial \theta} \right)^3 + A_{-1}^{12} \frac{\partial u}{\partial x} \frac{\partial w}{\partial \theta} + B_{-1}^{12} \frac{\partial \psi^x}{\partial x} \frac{\partial w}{\partial \theta} \\
& + \left( \frac{A_{-1}^{12}}{2} + A_{-1}^{66} \right) \frac{\partial w}{\partial \theta} \left( \frac{\partial w}{\partial x} \right)^2 + A_0^{66} \frac{\partial v}{\partial x} \frac{\partial w}{\partial x} \\
& + B_0^{66} \frac{\partial \psi^\theta}{\partial x} \frac{\partial w}{\partial x} + A_{-1}^{66} \frac{\partial u}{\partial \theta} \frac{\partial w}{\partial x} + B_{-1}^{66} \frac{\partial \psi^x}{\partial \theta} \frac{\partial w}{\partial x} \Big|_{\theta=2\pi}, \\
& \left[ A_0^{26} \frac{\partial v}{\partial x} + A_{-1}^{26} \frac{\partial u}{\partial \theta} + B_{-1}^{16} \frac{\partial \psi^x}{\partial \theta} + B_0^{26} \frac{\partial \psi^\theta}{\partial x} \right. \\
& \left. + \frac{A_0^{12}}{2} \left( \frac{\partial w}{\partial x} \right)^2 + \frac{A_{-2}^{22}}{2} \left( \frac{\partial w}{\partial \theta} \right)^2 + A_{-1}^{26} \frac{\partial w}{\partial x} \frac{\partial w}{\partial \theta} \right] \Big|_{\theta=0} = \\
& \left[ A_0^{26} \frac{\partial v}{\partial x} + A_{-1}^{26} \frac{\partial u}{\partial \theta} + B_{-1}^{16} \frac{\partial \psi^x}{\partial \theta} + B_0^{26} \frac{\partial \psi^\theta}{\partial x} \right. \\
& \left. + \frac{A_0^{12}}{2} \left( \frac{\partial w}{\partial x} \right)^2 + \frac{A_{-2}^{22}}{2} \left( \frac{\partial w}{\partial \theta} \right)^2 + A_{-1}^{26} \frac{\partial w}{\partial x} \frac{\partial w}{\partial \theta} \right] \Big|_{\theta=2\pi}, \\
& \left[ B_0^{26} \frac{\partial v}{\partial x} + B_{-1}^{26} \frac{\partial u}{\partial \theta} + D_{-1}^{16} \frac{\partial \psi^x}{\partial \theta} + D_0^{26} \frac{\partial \psi^\theta}{\partial x} + \frac{B_0^{12}}{2} \left( \frac{\partial w}{\partial x} \right)^2 \right. \\
& \left. + \frac{B_{-2}^{22}}{2} \left( \frac{\partial w}{\partial \theta} \right)^2 + B_{-1}^{26} \frac{\partial w}{\partial x} \frac{\partial w}{\partial \theta} \right] \Big|_{\theta=0} = \\
& \left[ B_0^{26} \frac{\partial v}{\partial x} + B_{-1}^{26} \frac{\partial u}{\partial \theta} + D_{-1}^{16} \frac{\partial \psi^x}{\partial \theta} + D_0^{26} \frac{\partial \psi^\theta}{\partial x} + \frac{B_0^{12}}{2} \left( \frac{\partial w}{\partial x} \right)^2 \right. \\
& \left. + \frac{B_{-2}^{22}}{2} \left( \frac{\partial w}{\partial \theta} \right)^2 + B_{-1}^{26} \frac{\partial w}{\partial x} \frac{\partial w}{\partial \theta} \right] \Big|_{\theta=2\pi}
\end{aligned} \quad (11a-j)$$

where

$$\begin{aligned}
& \left[ A_k^{ij} \quad B_k^{ij} \quad D_k^{ij} \quad L_k^{ij} \right] = \sum_{e=1}^{N_e} \int_{z_1^{(e)}}^{z_2^{(e)}} \bar{Q}_{ij}^{(e)} [1 \quad z \quad z^2 \quad z^3] (R+z)^k dz, \\
& \text{for } i, j = 1, 2, \dots, 6 \text{ and } k = -1, 0, 1 \\
& (I_{11}, I_{12}, I_{13}) = \sum_{e=1}^{N_e} \int_{z_1^{(e)}}^{z_2^{(e)}} \rho^{(e)} (1, z, z^2) (R+z) dz
\end{aligned} \quad (12a,b)$$

In the above Eq. (6)–(11),  $\bar{Q}_{ij}^{(e)}$  ( $i, j = 1, 2, \dots, 6$ ) are the transformed components of the layer stiffness [73];  $N_e$  is the number of the layers;  $z_1^{(e)}$  and  $z_2^{(e)}$  are the thickness coordinate of the inner and outer surfaces of the  $e$ th GRE laminated composite shell layer, respectively and  $H^\alpha$  ( $\alpha = x, \theta$ ) is the Heaviside function.

In this study, clamped boundary conditions at both ends  $x = 0$  and  $x = L$  of the GRE composite shell are considered as follows:

$$u = 0, v = 0, w = 0, \psi^x = 0, \psi^\theta = 0 \quad (13a-e)$$

and the initial conditions are assumed to be, respectively

$$u(x, \theta, z, 0) = 0, v(x, \theta, z, 0) = 0, w(x, \theta, z, 0) = 0,$$

$$\psi^x(x, \theta, z, 0) = 0, \psi^\theta(x, \theta, z, 0) = 0$$

$$\begin{aligned}
& \frac{\partial u(x, \theta, z, t)}{\partial t} \Big|_{t=0} = 0, \frac{\partial v(x, \theta, z, t)}{\partial t} \Big|_{t=0} = 0, \\
& \frac{\partial w(x, \theta, z, t)}{\partial t} \Big|_{t=0} = 0, \frac{\partial \psi^x(x, \theta, z, t)}{\partial t} \Big|_{t=0} = 0, \\
& \frac{\partial \psi^\theta(x, \theta, z, t)}{\partial t} \Big|_{t=0} = 0
\end{aligned} \quad (14a-j)$$

### 3. Solution procedure

Due to the complexity of the nonlinear governing differential equations, providing an analytical closed-form solution is barely achievable. Hence, an appropriate approximate analytical or numerical method should be used to solve them. DQM is opted to tackle this problem on, in which its computational efficiency and accuracy for different structural problems have been already well established [61–71]. However,

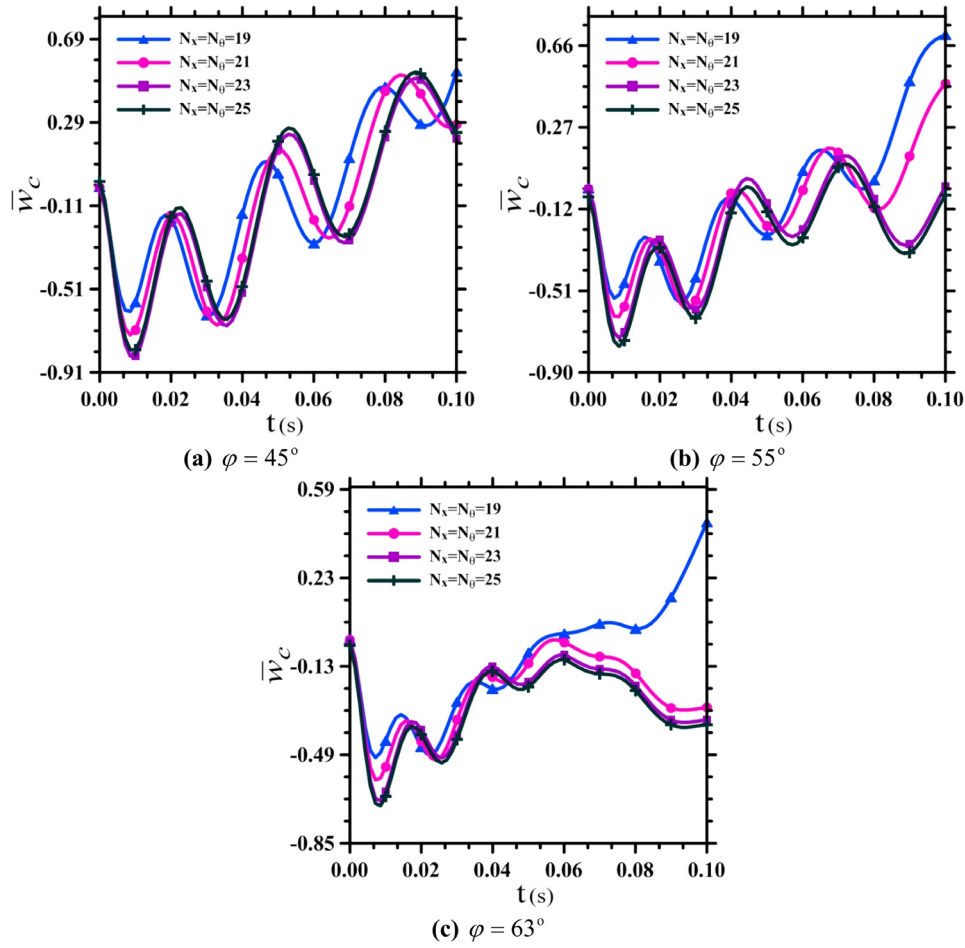


Fig. 2. (a)–(c): Convergence of the centre deflection against the number of DQ grid points for the  $[\varphi/\varphi]$  GRE laminated composite cylindrical shells  $[h/R = 0.01, X_0 = \Theta_0 = 0.5, P = 200 \text{ (kN)}, L/R = 4, N_L = 2, t^* = 0.004 \text{ (s)}]$ .

this method may encounter some difficulties when used to model the problems involving singular functions such as the Dirac-delta function. This is caused by the fact that the DQM cannot be directly employed to discretize the Dirac-delta function. To overcome this difficulty, in the current work, the DQM based on the direct projection of the Heaviside function [72] was utilized to discretize the governing equations in the spatial domain (detailed in Sections 1–3). Then, the resulting system of differential equations are solved in time domain using NURBS based on a multi-step technique [64,69] (detailed in Sections 2–3).

### 3.1. Spatial discretization: DQM-Heaviside function

To employ the DQM, the GRE composite shells are discretized into a set of  $N_x$  and  $N_\theta$  grid points along the  $x$ - and  $\theta$ -directions, respectively using the cosine-type grid generation rule [61–71]. The spatial derivatives in the differential equations of motion (6)–(10) together with the corresponding boundary and compatibility conditions at the interface of different layers (11), (13) are then discretized. The DQ discretized form of Eq. (6) is only presented here for the sake of brevity:

$$\begin{aligned} & A_1^{11} \sum_{k=1}^{N_x} B_{ik}^x u_{kj} + A_1^{16} \sum_{k=1}^{N_x} B_{ik}^x v_{kj} + A_{-1}^{66} \sum_{k=1}^{N_\theta} B_{jk}^\theta u_{ik} \\ & + (A_0^{66} + A_0^{12}) \sum_{k=1}^{N_x} \sum_{l=1}^{N_\theta} A_{ik}^x A_{jl}^\theta v_{kl} + 2A_0^{16} \sum_{k=1}^{N_x} \sum_{l=1}^{N_\theta} A_{ik}^x A_{jl}^\theta u_{kl} \\ & + A_0^{12} \sum_{k=1}^{N_x} A_{ik}^x w_{kj} + B_1^{11} \sum_{k=1}^{N_x} B_{ik}^x \psi_{kj} + B_{-1}^{16} \sum_{k=1}^{N_\theta} B_{jk}^\theta \psi_{ik} \end{aligned}$$

$$\begin{aligned} & + 2B_0^{16} \sum_{k=1}^{N_x} \sum_{l=1}^{N_\theta} A_{ik}^x A_{jl}^\theta \psi_{kl} + (B_0^{66} + B_0^{12}) \sum_{k=1}^{N_x} \sum_{l=1}^{N_\theta} A_{ik}^x A_{jl}^\theta \psi_{kl} + A_{-1}^{26} \sum_{k=1}^{N_\theta} B_{jk}^\theta v_{ik} \\ & + A_{-1}^{26} \sum_{k=1}^{N_\theta} B_{jk}^\theta v_{ik} + B_{-1}^{26} \sum_{k=1}^{N_\theta} A_{jk}^\theta w_{ik} + A_{-1}^{11} \left( \sum_{k=1}^{N_x} A_{ik}^x w_{kj} \right) \left( \sum_{k=1}^{N_x} B_{ik}^x w_{kj} \right) \\ & + (A_{-1}^{12} + A_{-1}^{66}) \left( \sum_{k=1}^{N_x} \sum_{l=1}^{N_\theta} A_{ik}^x A_{jl}^\theta w_{kl} \right) \left( \sum_{k=1}^{N_\theta} A_{jk}^\theta w_{ik} \right) \\ & + A_{-1}^{66} \left( \sum_{k=1}^{N_\theta} B_{jk}^\theta w_{ik} \right) \left( \sum_{k=1}^{N_x} A_{ik}^x w_{kj} \right) + A_0^{16} \left( \sum_{k=1}^{N_x} B_{ik}^x w_{kj} \right) \left( \sum_{k=1}^{N_\theta} A_{jk}^\theta w_{ik} \right) \\ & + 2A_0^{16} \left( \sum_{k=1}^{N_x} \sum_{l=1}^{N_\theta} A_{ik}^x A_{jl}^\theta w_{kl} \right) \left( \sum_{k=1}^{N_x} A_{ik}^x w_{kj} \right) = I_{11} \frac{d^2 u_{ij}}{dt^2} + I_{12} \frac{d^2 \psi_{ij}^x}{dt^2} \quad (15) \end{aligned}$$

where  $i = 2, \dots, N_x - 1$ ;  $j = 2, \dots, N_\theta - 1$ ;  $A_{ij}^\theta$  and  $B_{ij}^\theta$  ( $\beta = x, \theta$ ) denote, respectively, the first- and the second-order DQ weighting coefficients corresponding to the  $\beta$ -direction [61–71].

### 3.2. Temporal discretization: the NURBS based multi-step method

After completing the DQM discretization procedure in conjunction with the Heaviside function, one gets a system of ordinary differential equations (ODEs) in the temporal domain as

$$[M] \frac{d^2}{dt^2} \{D\} + [K] \{D\} = \{f(t)\} \quad (16)$$

where  $\{D\}$  indicates the vector of degrees of freedom;  $[M]$  and  $[K]$  are the mass and stiffness matrices, respectively, and  $\{f(t)\}$  is the load vector.



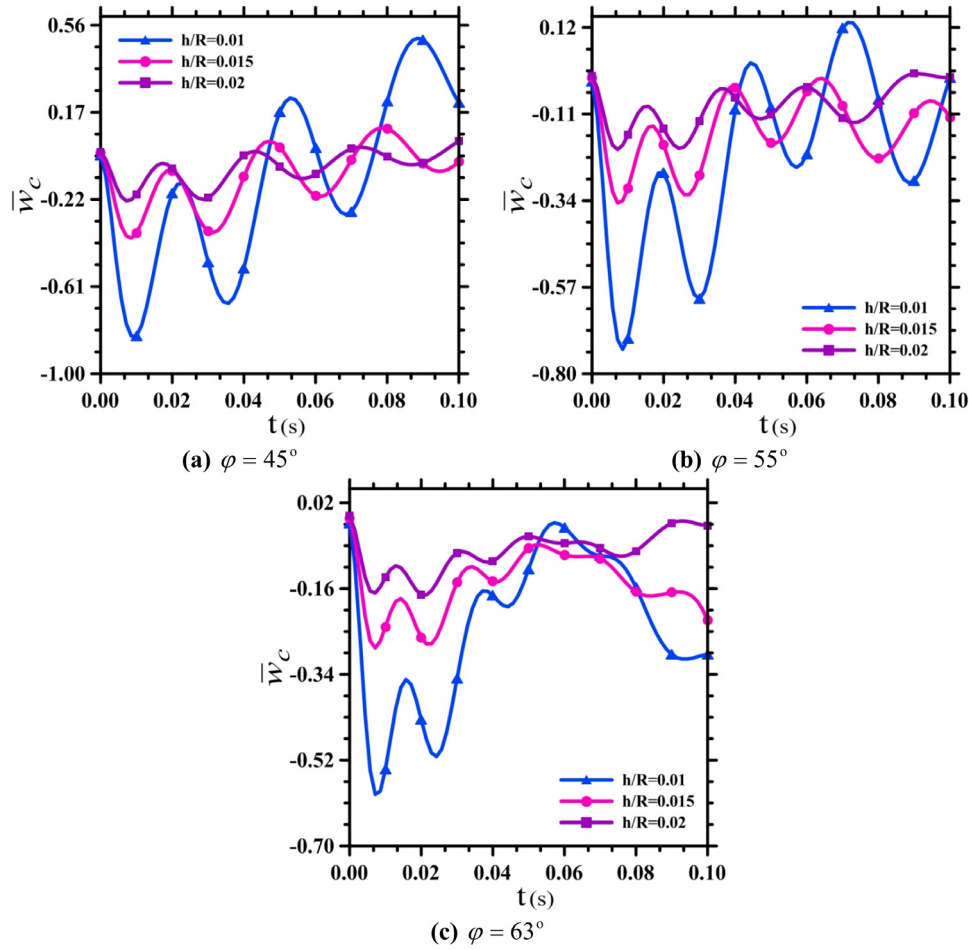


Fig. 3. (a)–(c): The effect of the thickness-to-mean radius ratio on the centre deflection of the  $[\varphi/-\varphi]$  GRE laminated composite cylindrical shells  $[X_0 = \Theta_0 = 0.5, P = 200 \text{ (kN)}, L/R = 4, N_L = 2, t^* = 0.004 \text{ (s)}]$ .

A multi-step method based on the NURBS curves [64,69] is used to solve the system of the ordinary differential equations (16) subjected to the related initial conditions. The superiority of this technique over the conventional finite difference-based schemes, such as Newmark's time integration scheme, has been confirmed by Heydarpour and Aghdam [64,69]. To apply this technique, the system of second-order ODEs (16) are transformed into a system of the first-order ones as

$$\begin{cases} \frac{d}{dt} \{\chi_1\} = \{\chi_2\} \\ [M] \frac{d}{dt} \{\chi_2\} + [K] \{\chi_1\} = \{f(t)\} \end{cases} \quad (17a,b)$$

where  $\{\chi_1\} = \{D\}$  and  $\{\chi_2\} = \frac{d}{dt} \{D\}$ . In employing the present approach, the different multi-step schemes can be obtained by changing the order and/or the weighting coefficients  $w_i$  of NURBS curves. In the present analysis, a four-step formula [64,69] with the weight coefficients  $w_1 = 0.001$ ,  $w_2 = 0.001$ ,  $w_3 = 2$  and  $w_4 = 3$  is utilized to discretize Eqs. (17a,b) to (18), (19), respectively:

$$\begin{aligned} \{\chi_1\}_{n+1} = & \{\chi_1\}_n + \Delta t (1.50002585 \{\chi_2\}_n - 0.50005291 \{\chi_2\}_{n-1} \\ & + 0.00002827 \{\chi_2\}_{n-2} - 0.00000120 \{\chi_2\}_{n-3}) \end{aligned} \quad (18)$$

$$\begin{aligned} \{\chi_2\}_{n+1} = & \{\chi_2\}_n + \Delta t (1.50002585 \{\hat{\chi}\}_n - 0.50005291 \{\hat{\chi}\}_{n-1} \\ & + 0.00002827 \{\hat{\chi}\}_{n-2} - 0.00000120 \{\hat{\chi}\}_{n-3}) \end{aligned} \quad (19)$$

where  $\Delta t$  is the time step size and

$$\{\hat{\chi}\} = [M]^{-1} (\{f(t)\} - [K] \{\chi_1\}) \quad (20)$$

To calculate the values of the unknown field variables in the first four points, the solution process starts by using the initial condition.

It should be noted that the different steps of the multi-step formulas are employed to obtain the data of the next three points. Subsequently, by performing the multi-step scheme, Eqs. (17a,b) are converted into a system of nonlinear algebraic equations in which their solution yields the values of unknown field variables at the end of each time step. Furthermore, an incremental-iterative method should be employed to solve the resulting nonlinear system of equations. In this study, the solution algorithms are based on the Newton–Raphson approach. Consequently, by reiterating the process, the displacement components at the DQ grid points of the GRE laminated composite cylindrical shell are achieved.

#### 4. Numerical results

In this section, the presented approach is validated, and then parametric studies are demonstrated. In the solved examples,  $\bar{w}_c = w(0.5, 0.5, t)/h$  is the non-dimensional centre radial displacement (deflection). Also, the material properties are chosen from [52], which are defined as

$$E_{11} = 44.5 \text{ (GPa)}, E_{22} = 12.2 \text{ (GPa)}, G_{12} = 4.33 \text{ (GPa)},$$

$$\rho = 2009.6 \text{ (kg/m}^3\text{)}, \nu_{12} = 0.28.$$

Usually,  $k_s = 5/(6 - \nu_{12})$  is used for composite materials as a shear correction factor in Eq. (12a).

As a first example, the rate of convergence of NURBS-based multi-step for nonlinear centre deflection of the GRE laminated composite cylindrical shells in the time domain is presented in Table 1. The results are obtained for three different winding angles and two different sets of times. The fast rate of convergence of the NURBS scheme is quite obvious in this table.

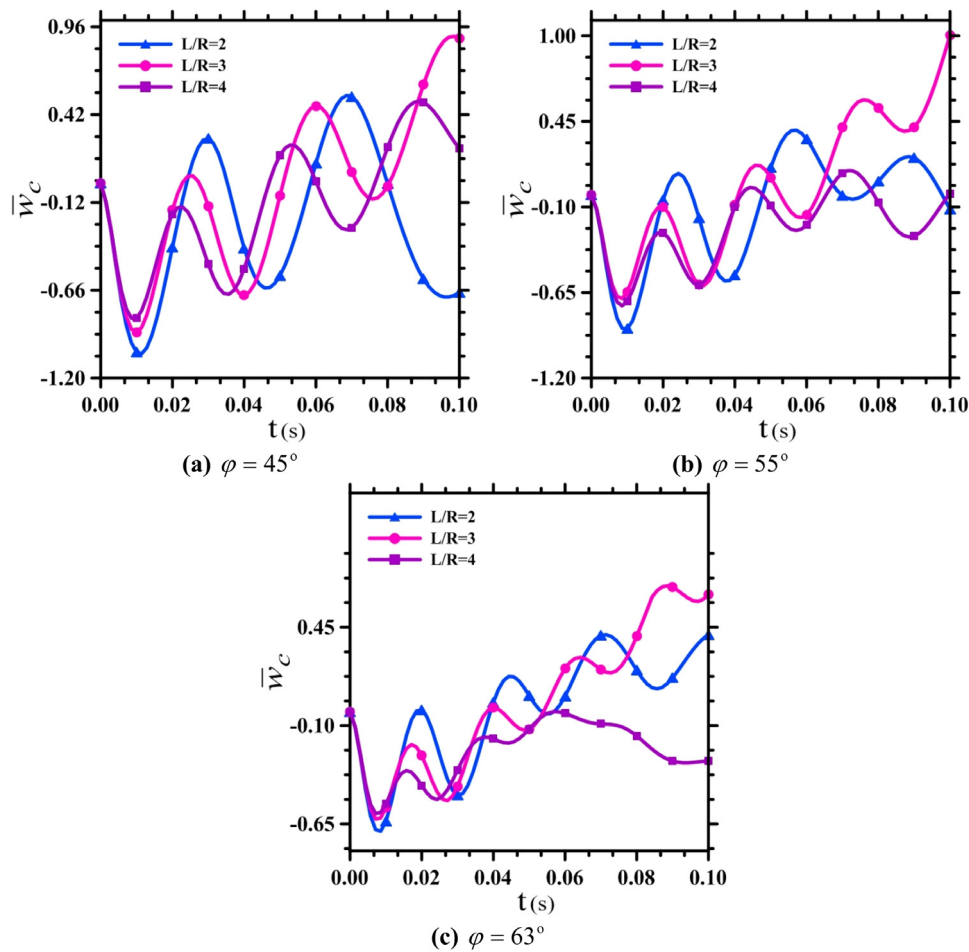


Fig. 4. (a)–(c): The effect of the length-to-mean radius ratio on the centre deflection of the  $[\varphi/-\varphi]$  GRE laminated composite cylindrical shells  $[X_0 = \Theta_0 = 0.5, P = 200 \text{ (kN)}, h/R = 0.01, N_L = 2, t^* = 0.004 \text{ (s)}]$ .

Table 1

Convergence of the centre deflections of the  $[\varphi/-\varphi]$  GRE laminated composite cylindrical shells  $[h/R = 0.01, X_0 = \Theta_0 = 0.5, P = 200 \text{ (kN)}, L/R = 4, N_L = 2, t^* = 0.004 \text{ (s)}]$ .

| $N_t$ | $t = 0.03 \text{ (s)}$ |                      |                      | $t = 0.06 \text{ (s)}$ |                      |                      |
|-------|------------------------|----------------------|----------------------|------------------------|----------------------|----------------------|
|       | $\varphi = 45^\circ$   | $\varphi = 55^\circ$ | $\varphi = 63^\circ$ | $\varphi = 45^\circ$   | $\varphi = 55^\circ$ | $\varphi = 63^\circ$ |
| 100   | -4.5095                | -5.8641              | -7.8746              | 3.6598                 | -6.5060              | -5.5317              |
| 400   | -1.0121                | -2.4381              | -1.8941              | 0.2617                 | -1.3752              | -2.6042              |
| 800   | -0.7654                | -0.9027              | -0.7581              | 0.0874                 | -0.8967              | -0.8490              |
| 1000  | -0.5127                | -0.6201              | -0.4219              | 0.0129                 | -0.2407              | -0.1183              |
| 1200  | -0.5005                | -0.6010              | -0.4091              | 0.0098                 | -0.2174              | -0.0824              |
| 1500  | -0.5001                | -0.6004              | -0.4080              | 0.0095                 | -0.2167              | -0.0817              |

Table 2 compares CPU time needed for the nonlinear dynamic analysis of a GRE laminated composite cylindrical shell subjected to the impulse loading using the NURBS-based multi-step time integration scheme and Newmark's method. The rate of convergence of the non-dimensional centre deflection versus the number of time steps ( $N_t$ ) is presented. The less CPU time of the NURBS-based multi-step technique over the Newmark method is quite noticeable in this table. For example, for  $N_t=1200$ , the computational cost of the NURBS-based multistep method is 19.82095 s while the CPU time of Newmark scheme is 1864.912087 s to reach a level of convergence up to 3 digits. This implies that for a similar accuracy, the presented method requires about 100 times smaller CPU time to provide appropriate convergence.

The convergence rate of the proposed method for the nonlinear dynamic analysis of GRE laminated composite cylindrical shells under

Table 2

Comparison between NURBS-based multi-step method and Newmark approach for the dynamic response of the  $[\varphi/-\varphi]$  GRE composite shells  $[h/R = 0.01, X_0 = \Theta_0 = 0.5, P = 200 \text{ (kN)}, L/R = 4, N_L = 2, \varphi = 55^\circ, t = 0.05 \text{ s}, t^* = 0.008 \text{ (s)}]$ .

| $N_t$ | NURBS-based scheme |              | Newmark's scheme |              |
|-------|--------------------|--------------|------------------|--------------|
|       | $\bar{w}_c$        | CPU time (s) | $\bar{w}_c$      | CPU time (s) |
| 100   | 3.7950             | 1.155907     | 1.8720           | 184.069472   |
| 150   | 1.3061             | 1.948201     | 0.6167           | 266.701265   |
| 200   | 0.4594             | 3.503768     | 0.2208           | 398.615704   |
| 400   | 0.2104             | 5.936477     | 0.1649           | 787.448607   |
| 600   | 0.1563             | 7.007649     | 0.1486           | 987.500810   |
| 800   | 0.1487             | 11.86031     | 0.1453           | 1364.820497  |
| 1000  | 0.1450             | 14.02574     | 0.1450           | 1593.226482  |
| 1200  | 0.1454             | 19.82095     | 0.1449           | 1864.912087  |

impulse loading is shown in Fig. 2(a)–(c). In these figures, the time history of the dimensionless deflection of the centre against the DQ number of grid points in the thickness direction ( $N_x$ ) and circumferential direction ( $N_\theta$ ) is shown. One can observe that twenty-five grid points (i.e.  $N_x = N_\theta = 25$ ) in the axial and circumferential directions yield fairly acceptable results. Hence, the values of  $N_x = N_\theta = 25$  and  $N_t = 1200$  are chosen as the optimum values to generate all other numerical results.

To validate the presented model and solution methodology, the natural frequencies of a simply supported FG cylindrical shell are extracted and compared with the ones presented by Loy et al. [74]. They used Love's shell theory in conjunction with the Rayleigh–Ritz

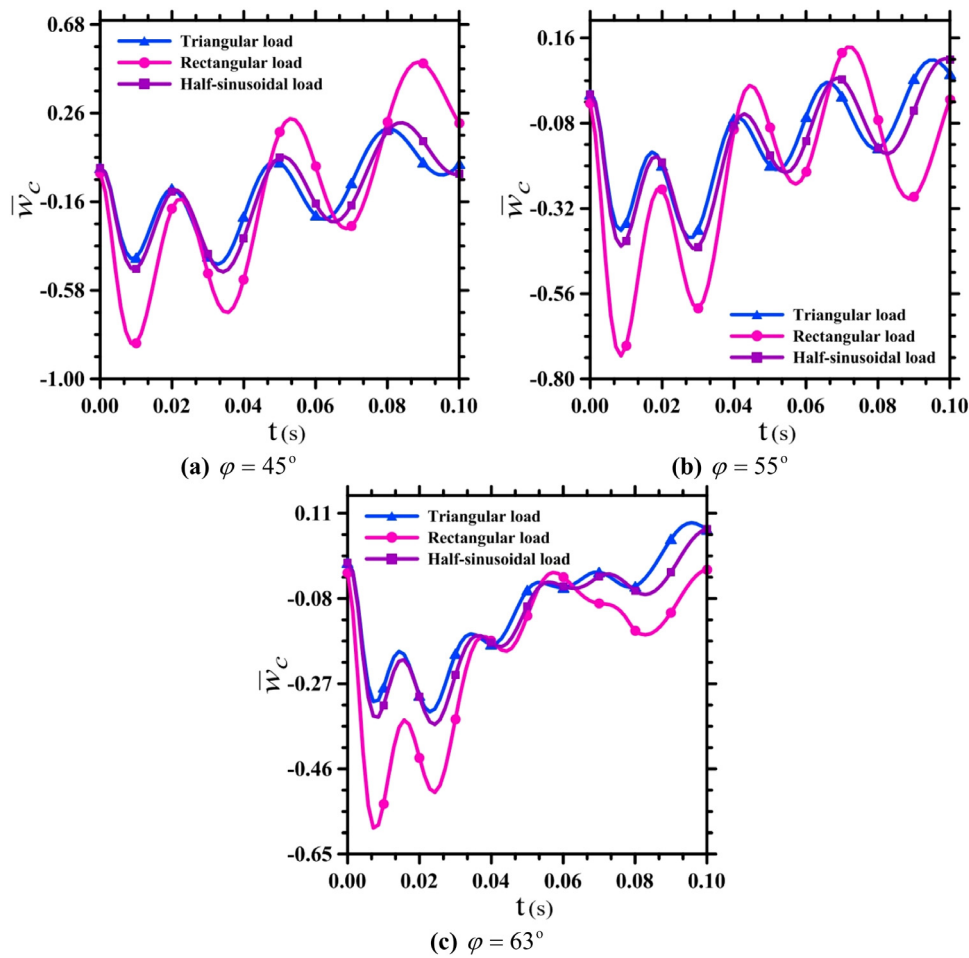


Fig. 5. (a)–(c): The effect of the three different types of impulse loading on the centre deflection of the  $[\varphi / -\varphi]$  GRE laminated composite cylindrical shells  $[X_0 = \Theta_0 = 0.5, P = 200 \text{ (kN)}, h/R = 0.01, L/R = 4, N_L = 2, t^* = 0.004 \text{ (s)}]$ .

**Table 3**  
Material properties of the metal and ceramic [74].

| Material                    | Metal   | Ceramic |
|-----------------------------|---------|---------|
| $E$ (GPa)                   | 207.788 | 205.098 |
| $\nu$                       | 0.318   | 0.31    |
| $\rho$ (kg/m <sup>3</sup> ) | 8166    | 8900    |

method to analyse the free vibration of FG cylindrical shells. Moreover, the material properties vary through the shell thickness from ceramic-rich at the inner surface to metal-rich at the outer surface according to the power-law distribution of

$$P(z) = P_C + (P_M - P_C) \left( \frac{2z + h}{2h} \right)^p \quad (21)$$

where  $P$  is a generic material property and the subscripts  $M$  and  $C$  refer to the metal and ceramic constituents, respectively;  $p$  is a positive real number known as the power law index (or the material property gradient index). The metal and ceramic properties are listed in Table 3. Table 4 compares the natural frequencies associated with the first three circumferential modes for the three different values of the material graded-index  $p$ , which clearly demonstrates a close agreement between two approaches. However, some slight differences between the present results and Ref. [74] might be due to the different shell theories and method of solution. For instance, based on Love's shell theory  $\sigma_{zx}$  and  $\sigma_{z\theta}$  are ignored whereas these two stress tensor components are considered in FSDT theory.

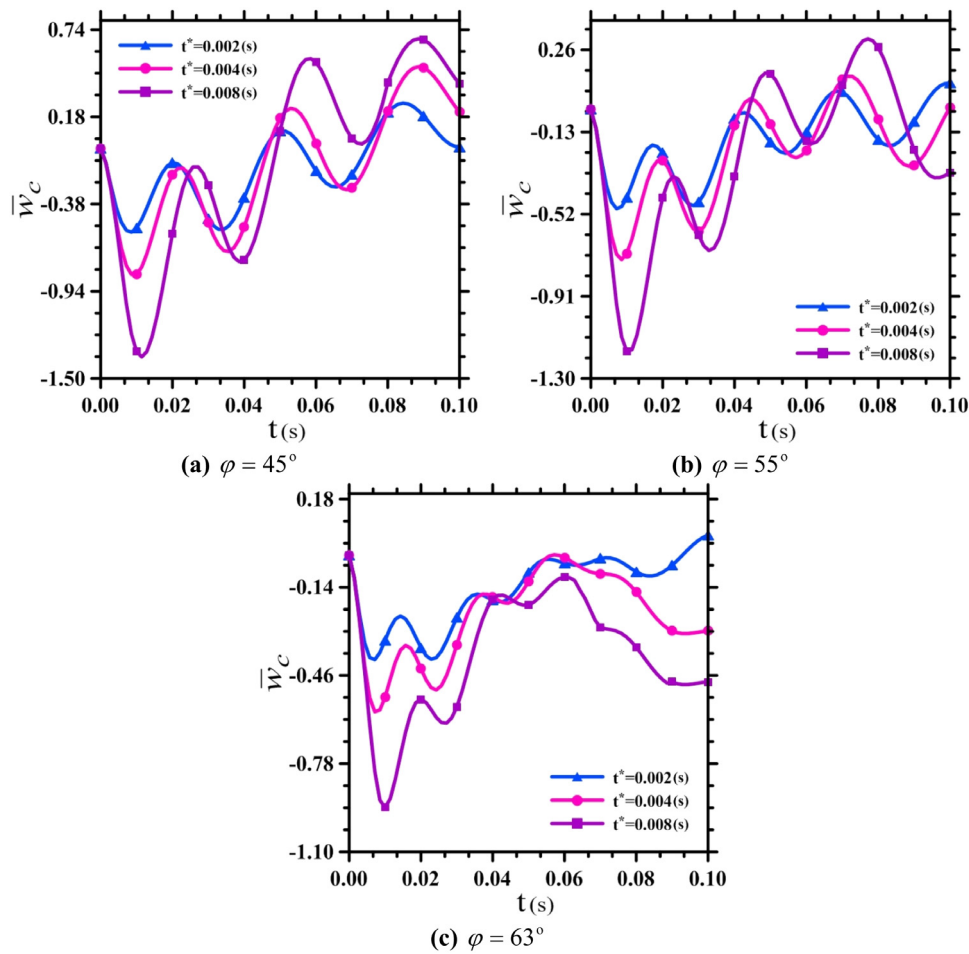
**Table 4**  
Comparison of the natural frequencies (Hz) for a soft simply supported FG cylindrical shell ( $h/R = 0.05, L/R = 20$ ).

| $p$ | $m$ | Proposed method | Loy et al. [74] | Percent different |
|-----|-----|-----------------|-----------------|-------------------|
| 0   | 1   | 12.898          | 12.917          | 0.147%            |
|     | 2   | 31.543          | 31.603          | 0.190%            |
|     | 3   | 87.977          | 88.267          | 0.330%            |
| 1   | 1   | 13.195          | 13.234          | 0.296%            |
|     | 2   | 32.269          | 32.418          | 0.462%            |
|     | 3   | 90.007          | 90.569          | 0.624%            |
| 2   | 1   | 13.305          | 13.344          | 0.293%            |
|     | 2   | 32.514          | 32.683          | 0.520%            |
|     | 3   | 90.689          | 91.309          | 0.684%            |

As another comparative study, the nonlinear frequency ratios ( $\omega_{NL}/\omega_L$ ) are compared with the method of multiple scales reported by Sheng and Wang [75] in Table 5. They studied the nonlinear dynamic behaviours of FG cylindrical shells under combined parametric and external excitations based on Hamilton's principle and the first-order shear deformation theory. For the purpose of comparison, a homogeneous orthotropic cylindrical shell with simply supported boundary conditions is considered. The material properties and geometrical parameters are as follows [75]

$$E_{11} = 200 \text{ (GPa)}, E_{22} = G_{12} = 10 \text{ (GPa)}, G_{13} = G_{23} = 0.8G_{12}, \\ \rho = 7800 \text{ (kg/m}^3\text{)}, \nu_{12} = 0.2, \\ L = 0.4 \text{ (m)}, h = 0.00245 \text{ (m)}, R = 0.245 \text{ (m)}.$$





**Fig. 6.** (a)–(c): The effect of the three different time durations of impulse loading on the centre deflection of the  $[\varphi / -\varphi]$  GRE laminated composite cylindrical shells  $[X_0 = \Theta_0 = 0.5, P = 200 \text{ (kN)}, h/R = 0.01, L/R = 4, N_L = 2]$ .

**Table 5**

Comparison of the nonlinear to linear frequency ratio ( $\omega_{NL}/\omega_L$ ) for homogeneous orthotropic cylindrical shell.

| $a_l/h$ | Present | Sheng and Wang [75] |
|---------|---------|---------------------|
| 0.5     | 1.00174 | 1.0017              |
| 1       | 1.00713 | 1.0066              |
| 1.5     | 1.01638 | 1.0149              |
| 2       | 1.02807 | 1.0265              |
| 2.5     | 1.04356 | 1.0414              |

Table 5 indicates the fact that, in all cases, the close agreements are observed between the present result and the result of Sheng and Wang [75].

The effect of the thickness-to-mean radius ratio of the shell on the nonlinear dynamic behaviour of GRE laminated composite cylindrical shells under impulse loading is investigated in Fig. 3(a)–(c). The results are reported for three different values of the winding angles. As expected, the nonlinear centre deflection decreases with increasing the thickness-to-mean radius ratio. This is due to the increase of the shell stiffness as the thickness-to-mean radius ratio rises.

Fig. 4(a)–(c) depict the effects of the length-to-mean radius ratio on the time history of the dimensionless nonlinear centre deflection of the GRE laminated composite cylindrical shells at three different winding angles  $\varphi = 45^\circ, 55^\circ, 63^\circ$ . The change of length-to-mean radius ratio has a considerable effect on the time histories of the centre deflection as it is clear in these figures. It is also observed that with increasing this parameter, the shell stiffness increases, and consequently, the centre deflection decreases. Moreover, with increasing  $\varphi$ , the nonlinear centre

deflection of the GRE composite shells declines. As reported in previous studies [4], the strength of the GRE laminated composite cylindrical shells decreases as the winding angle increases. This could be probably due to the orientation of the glass fibre during winding to the specified angle. The period of oscillatory portions of the deflection can be easily seen from Fig. 4. The period of oscillatory portions of the deflection in the shell with  $L/R = 2$  is lower than the shells with other length-to-mean radius ratios and as this parameter increases, the period of oscillatory portions of the deflection increases too.

The time history of the nonlinear centre deflection of the GRE laminated composite cylindrical shells against the three different types of impulse loading is plotted in Fig. 5(a)–(c). In these figures, the time history of the centre deflection is plotted for triangular, rectangular, and half-sinusoidal types of impulse loads. The results show that the effects of the triangular and half-sinusoidal types of impulse loading on the nonlinear centre deflection are same, while the rectangular impulse load type has a stronger effect on the deflection responses. This is because the area beneath the rectangular load time history is larger than the other two types of impact loading.

The effect of the three different time durations ( $t^*$ ) of impulse loading on the nonlinear dynamic behaviour of the GRE laminated composite cylindrical shells is investigated by showing the time history of the centre deflection in Fig. 6. Significant differences are observed between the results obtained by different values of time durations of impulse loading. The growth on the nonlinear deflection of the GRE composite shells for the impulse loadings with the longer time durations ( $t^*$ ) is obvious from Fig. 6. In addition, an increase in the time durations of impulse loading does not affect the period of the motion in the oscillatory parts of the results, as is clearly revealed in this figure.

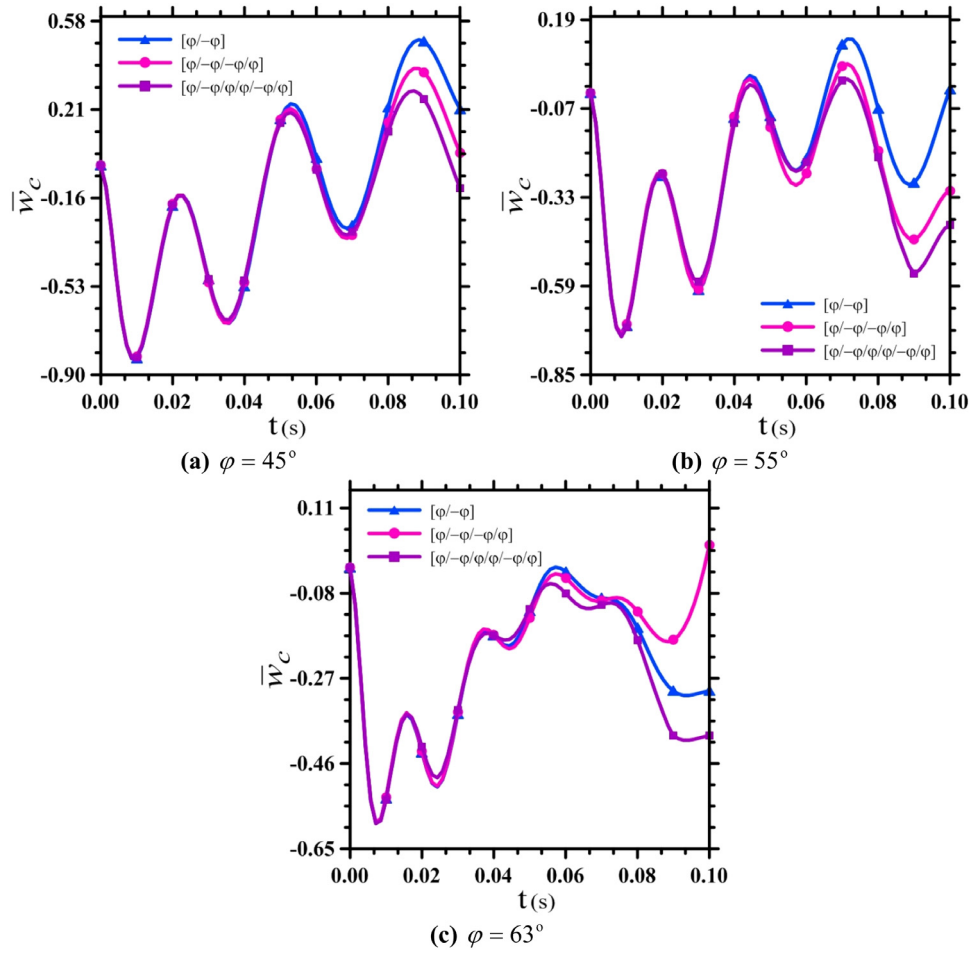


Fig. 7. (a)–(c): The effect of number of layers on the centre deflection of the GRE laminated composite cylindrical shells [ $X_0 = \Theta_0 = 0.5$ ,  $P = 200$  (kN),  $h/R = 0.01$ ,  $L/R = 4$ ,  $r^* = 0.004$  (s)].

The variations of the time histories of the nonlinear centre deflection of GRE laminated composite cylindrical shells under impulse loading at their outer surfaces against the number of layers are shown in Fig. 7. The total thickness of the shell is constant. As it is seen, number of the layers does not affect the time histories of the dynamic response of the shell at early times. However, with an increase in time ( $t$ ), the difference between the results increase for three different values of the winding angles. In these figures, by increasing the winding angles, the nonlinear centre deflection of the GRE composite shells decreases too. This clearly indicates that the overall stiffness of the GRE composite shell increases with increasing the winding angles.

Fig. 8 depict the impacts of different locations of the impulse loading on the nonlinear dynamic behaviour of GRE laminated composite cylindrical shells. In these figures, the time histories of the non-dimensional centre deflection for the four different locations of the impulse loading at three different values of the winding angles are presented. The location of the impulse loading represents a noticeable effect on the nonlinear centre deflection. The maximum centre deflection of the GRE composite shells under rectangular impulse loading occurs at  $X_0 = 0.4$ . Furthermore, it significantly changes the amplitude and period of oscillatory portions of the nonlinear deflection in the GRE laminated composite cylindrical shell.

## 5. Conclusions

For the first time, in this research, the first-order shear deformation theory of shells was employed to simulate the nonlinear dynamic behaviour in GRE laminated composite cylindrical shells subjected to

impulse loadings. Then a hybrid numerical method combined of (i) a differential quadrature method based on the direct projection of the Heaviside function (for the space domain) and (ii) a non-uniform rational B-spline (NURBS) based multi-step time integration scheme (for the time-domain) was proposed and employed to solve the equations. Furthermore, in order to solve the resulting nonlinear system of equations, the Newton–Raphson technique was used. The accuracy of the proposed method was demonstrated and justified through comparison with similar models in the literature. In addition, the proposed method was shown to outperform Newmark's technique by far in terms of CPU time. Then, the effects of the geometrical parameters, different time durations, and types of imposing loads and number of layers on the nonlinear dynamic behaviour of GRE laminated composite cylindrical shells at different values of the winding angles were studied. This comprehensive parametric study resulted in the following:

- The increase of the time durations of the impulse loading leads to a larger nonlinear centre deflection.
- As the thickness and length-to-mean radius ratios increase, the nonlinear centre deflection of the GRE composite shells decreases; whereas the period of oscillatory portions of the deflection increases.
- Increase of winding angle ( $\varphi$ ) results in an increase in the overall stiffness and reduces the deflection of the GRE composite shells.
- The location of the maximum values of the deflection is almost dependent on the value of the winding angle ( $\varphi$ ).
- The rectangular type of impulse loading has a more significant effect than the other types on the time histories of the deflection.

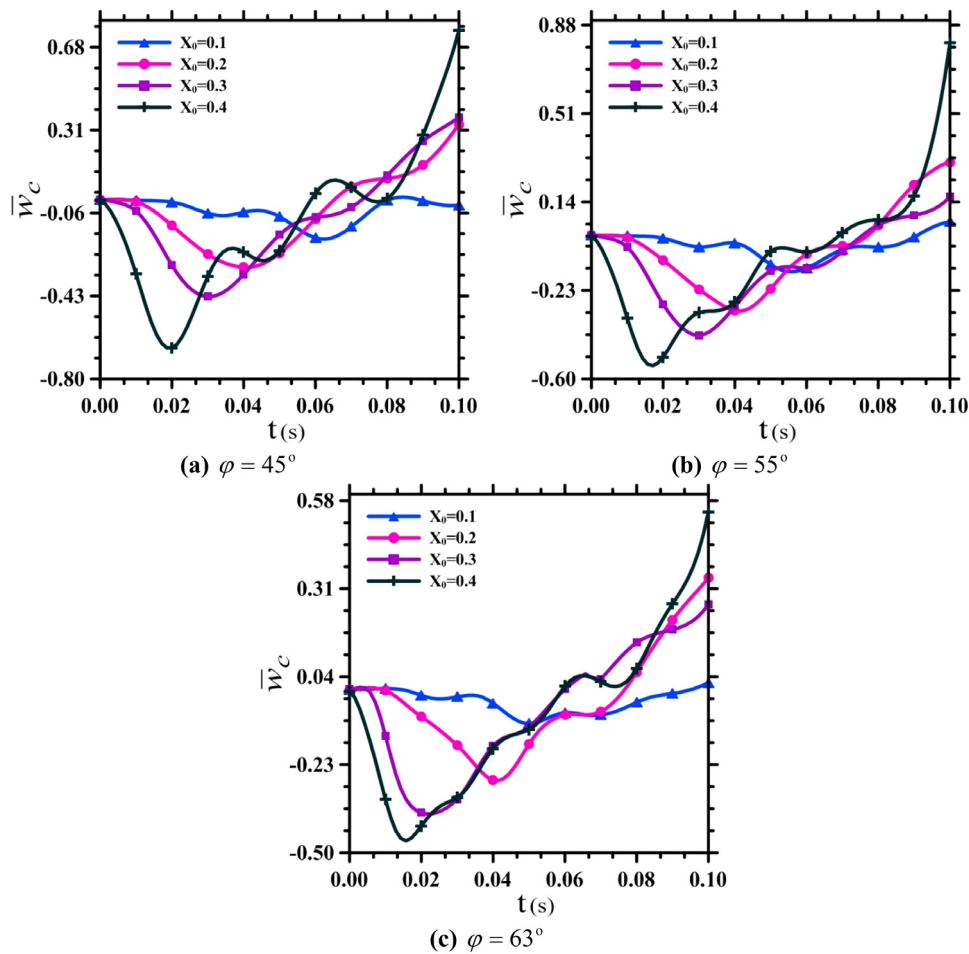


Fig. 8. (a)–(c): The effect of load location on the centre deflection of the GRE laminated composite cylindrical shells [ $\theta_0 = 0.5$ ,  $P = 200$  (kN),  $h/R = 0.01$ ,  $L/R = 4$ ,  $t^* = 0.004$  (s)].

- The number of layers has no effect on the nonlinear dynamic behaviour at early times.
- Load location noticeably influences the time history of the nonlinear deflection.

#### CRediT authorship contribution statement

**Yasin Heydarpour:** Programming, Formulation, Writing. **Morteza Mohammadzaheri:** Conceptualization, Supervision, Editing, Writing. **Mojtaba Ghodsi:** Reviewing and editing, Conceptualization. **Payam Soltani:** Reviewing and editing, Validation. **Farooq Al-Jahwari:** Reviewing and editing, Double checking with other methods. **Issam Bahadur:** Reviewing and editing. **Badar Al-Amri:** Data collection from industry sources.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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#### Appendix

Based on nonlinear Green's strain tensor considering von Kármán assumptions, the nonlinear strain displacement relations in the cylindrical coordinate system become [76]

$$\begin{aligned}
 \bar{\epsilon}_{xx} &= \epsilon_{xx}^L + \epsilon_{xx}^{NL} = \frac{\partial \bar{u}}{\partial x} + \frac{1}{2} \left[ \left( \frac{\partial \bar{u}}{\partial x} \right)^2 + \left( \frac{\partial \bar{v}}{\partial x} \right)^2 + \left( \frac{\partial \bar{w}}{\partial x} \right)^2 \right], \\
 \bar{\epsilon}_{\theta\theta} &= \epsilon_{\theta\theta}^L + \epsilon_{\theta\theta}^{NL} = \frac{1}{(R+z)} \left( \frac{\partial \bar{v}}{\partial \theta} + \bar{w} \right) \\
 &\quad + \frac{1}{2(R+z)^2} \left[ \left( \frac{\partial \bar{w}}{\partial \theta} - \bar{v} \right)^2 + \left( \frac{\partial \bar{v}}{\partial \theta} + \bar{w} \right)^2 + \left( \frac{\partial \bar{u}}{\partial \theta} \right)^2 \right], \\
 \bar{\epsilon}_{zz} &= \epsilon_{zz}^L + \epsilon_{zz}^{NL} = \frac{\partial \bar{w}}{\partial z} + \frac{1}{2} \left[ \left( \frac{\partial \bar{u}}{\partial z} \right)^2 + \left( \frac{\partial \bar{v}}{\partial z} \right)^2 + \left( \frac{\partial \bar{w}}{\partial z} \right)^2 \right], \\
 \bar{\gamma}_{z\theta} &= \gamma_{z\theta}^L + \gamma_{z\theta}^{NL} = \frac{\partial \bar{v}}{\partial z} + \frac{1}{(R+z)} \left( \frac{\partial \bar{w}}{\partial \theta} - \bar{v} \right) \\
 &\quad + \frac{1}{(R+z)} \left[ \frac{\partial \bar{w}}{\partial z} \left( \frac{\partial \bar{w}}{\partial \theta} - \bar{v} \right) + \frac{\partial \bar{v}}{\partial z} \left( \frac{\partial \bar{v}}{\partial \theta} + \bar{w} \right) + \frac{\partial \bar{u}}{\partial z} \frac{\partial \bar{u}}{\partial \theta} \right], \\
 \bar{\gamma}_{\theta x} &= \gamma_{\theta x}^L + \gamma_{\theta x}^{NL} = \frac{\partial \bar{v}}{\partial x} + \frac{\partial \bar{u}}{(R+z) \partial \theta} \\
 &\quad + \frac{1}{(R+z)} \left[ \frac{\partial \bar{w}}{\partial x} \left( \frac{\partial \bar{w}}{\partial \theta} - \bar{v} \right) + \frac{\partial \bar{v}}{\partial z} \left( \frac{\partial \bar{v}}{\partial \theta} + \bar{w} \right) + \frac{\partial \bar{u}}{\partial \theta} \frac{\partial \bar{u}}{\partial x} \right], \\
 \bar{\gamma}_{zx} &= \gamma_{zx}^L + \gamma_{zx}^{NL} = \frac{\partial \bar{w}}{\partial x} + \frac{\partial \bar{u}}{\partial z} + \frac{\partial \bar{u}}{\partial x} \frac{\partial \bar{u}}{\partial z} + \frac{\partial \bar{v}}{\partial x} \frac{\partial \bar{v}}{\partial z} + \frac{\partial \bar{w}}{\partial x} \frac{\partial \bar{w}}{\partial z}
 \end{aligned} \tag{A.1–6}$$

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